

268. Clearly, more fuel, the longer will the engine run.

$$\therefore \frac{4}{5} : \frac{1}{3} :: 12 : x \text{ or } \frac{4}{5}x = \frac{1}{3} \times 12 = 4 \text{ or } x = 4 \times \frac{5}{4} = 5.$$

269. Suppose the taller tree grows x feet in 1 year.

Then, the shorter tree grows $\frac{3x}{5}$ ft in 1 year.

$$\begin{aligned} \therefore 4x + 4 \times \frac{3}{5}x &= 8 \Leftrightarrow 4x + \frac{12}{5}x = 8 \Leftrightarrow \frac{32}{5}x = 8 \\ \Leftrightarrow x &= \frac{8 \times 5}{32} = \frac{5}{4}. \end{aligned}$$

$$\begin{aligned} \text{Growth of shorter tree in 2 years} &= \left(2 \times \frac{3x}{5}\right) \text{ ft} \\ &= \left(2 \times \frac{3}{5} \times \frac{5}{4}\right) \text{ ft} = \frac{3}{2} \text{ ft} = 1\frac{1}{2} \text{ ft}. \end{aligned}$$

270. Let the amount of fuel consumed while coming back be x gallons.

Then, amount of fuel consumed while going

$$= \left(x + \frac{x}{4}\right) \text{ gallons} = \frac{5x}{4} \text{ gallons}.$$

$$\therefore x + \frac{5x}{4} = 4\frac{1}{2} \Leftrightarrow \frac{9x}{4} = \frac{9}{2} \Leftrightarrow x = \frac{9}{2} \times \frac{4}{9} = 2.$$

271. Let the highest temperature be x degrees.

$$\begin{aligned} \text{Then, lowest temperature} &= \left[\left(1 + \frac{1}{3}\right)\frac{x}{2}\right] \text{ degrees} \\ &= \left(\frac{4}{3} \times \frac{x}{2}\right) \text{ degrees} = \frac{2x}{3} \text{ degrees}. \end{aligned}$$

$$\therefore x + \frac{2x}{3} = 100 \Leftrightarrow \frac{5x}{3} = 100 \Leftrightarrow x = \frac{100 \times 3}{5} = 60.$$

$$\text{So, lowest temperature} = \left(\frac{2}{3} \times 60\right) \text{ degrees} = 40 \text{ degrees}.$$

272. Let the total number of students in the class be x .

$$\text{Then, number of boys} = \frac{2x}{3}; \text{ number of girls} = \frac{x}{3}.$$

$$\text{Number of boys who are over 160 cm tall} = \left(\frac{3}{4} \times \frac{2x}{3}\right) = \frac{x}{2}.$$

$$\text{So, } \frac{x}{2} = 18 \text{ or } x = 36.$$

$$\therefore \text{Number of girls} = \frac{36}{3} = 12.$$

273. Suppose initially Peter had ₹ x . Then,

$$\text{Amount received by Michael} = ₹ \left(\frac{x}{4}\right). \text{ Amount remain-}$$

$$\text{ing with Peter} = ₹ \left(x - \frac{x}{4}\right) = ₹ \left(\frac{3x}{4}\right).$$

$$\text{Amount received by Sam} = ₹ \left(\frac{1}{2} \times \frac{x}{4}\right) = ₹ \left(\frac{x}{8}\right).$$

$$\therefore \frac{3x}{4} - \frac{x}{8} = 500 \Leftrightarrow 5x = 4000 \Leftrightarrow x = 800.$$

$$\text{Hence, amount received by Michael} = (x/4) = ₹ 200.$$

$$\mathbf{274.} \text{ A's share} = \frac{1}{3}. \text{ Remainder} = \left(1 - \frac{1}{3}\right) = \frac{2}{3}.$$

$$\text{B's share} = \frac{2}{5} \text{ of } \frac{2}{3} = \frac{4}{15}. \text{ Rest} = \left(\frac{2}{3} - \frac{4}{15}\right) = \frac{6}{15} = \frac{2}{5}.$$

$$\text{C's share} = \text{D's share} = \frac{1}{2} \text{ of } \frac{2}{5} = \frac{1}{5}.$$

275. Let the total length of the journey be x km.

$$\text{Then, } \frac{2}{5}x = 1200 \Leftrightarrow x = \frac{1200 \times 5}{2} = 3000.$$

$$\text{Distance travelled by car} = \left(\frac{1}{3} \times 3000\right) \text{ km} = 1000 \text{ km}.$$

$$\therefore \text{Distance travelled by train} = [3000 - (1200 + 1000)] \text{ km} = 800 \text{ km}.$$

276. Let the total income be ₹ x .

Then, remaining money

$$= ₹ \left[x - \left(\frac{x}{4} + \frac{x}{5}\right)\right] = ₹ \left(x - \frac{9x}{20}\right) = ₹ \frac{11x}{20}.$$

$$\text{So, } \frac{11x}{20} = 231 \Leftrightarrow x = \frac{231 \times 20}{11} = 420.$$

$$\mathbf{277.} \text{ Part read on first day} = \frac{3}{8}. \text{ Remaining part} = \left(1 - \frac{3}{8}\right) = \frac{5}{8}.$$

$$\text{Part read on second day} = \frac{4}{5} \text{ of } \frac{5}{8} = \frac{1}{2}.$$

$$\text{Unread part} = \left[1 - \left(\frac{3}{8} + \frac{1}{2}\right)\right] = \frac{1}{8}.$$

Let the number of pages be x .

$$\text{Then, } \frac{1}{8}x = 30 \text{ or } x = 30 \times 8 = 240.$$

278. Out of 5 girls, 1 took part in fete. Out of 8 boys, 1 took part in fete.

$\therefore$ Out of 13 students, 2 took part in fete.

$$\text{Hence, } \frac{2}{13} \text{ of the total number took part in fete}.$$

$$\mathbf{279.} \text{ French men} = \frac{1}{5}; \text{ French women} = \left(\frac{1}{5} + \frac{2}{3} \times \frac{1}{5}\right) = \frac{5}{15} = \frac{1}{3}.$$

$$\text{French people} = \left(\frac{1}{5} + \frac{1}{3}\right) = \frac{8}{15}.$$

$$\therefore \text{Not French} = \left(1 - \frac{8}{15}\right) = \frac{7}{15}.$$

$$\mathbf{280.} \text{ Girls} = \frac{3}{5}; \text{ Boys} = \left(1 - \frac{3}{5}\right) = \frac{2}{5}.$$

$$\begin{aligned} \text{Fraction of students absent} &= \frac{2}{9} \text{ of } \frac{3}{5} + \frac{1}{4} \text{ of } \frac{2}{5} \\ &= \frac{6}{45} + \frac{1}{10} = \frac{21}{90} = \frac{7}{30}. \end{aligned}$$

$$\therefore \text{Fraction of students present} = \left(1 - \frac{7}{30}\right) = \frac{23}{30}.$$

281. Number of boys who participate = 100.

$$\therefore \frac{1}{3} \text{ of boys} = 100 \text{ or total number of boys} = 300.$$